

Helix Thready — API Versioning Policy

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Rev	Date	Author	Change
1	2026-07-21	swarm (API & SDKs)	Initial draft: URL-path <code>/v1</code> , additive-vs-breaking, buf breaking gate
2	2026-07-21	swarm (API & SDKs)	Linked the breaking-change gate to its RED-first skeleton in contract-tests.md
3	2026-07-22	swarm (API & SDKs)	Closed <code>[OPEN: ver-2] : GET / version</code> (build + contract identity + active deprecations) is now in <code>openapi.yaml</code> (<code>getVersion</code> → <code>VersionInfo</code>).

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1. Decision

URL-path versioning: `/v1`. Resolves **Q29** `[DEFAULT — adjustable]`. Every REST route is mounted under `/v1` (`https://thready.hxd3v.com/v1/...`). The parallel Protobuf contract uses the matching package suffix `helix.thready.v1`, exactly as `helix_proto` pins `helix.<domain>.v1` `[RESEARCH: helix_proto]`. A future incompatible surface ships as `/v2` + `helix.thready.v2`, served **side-by-side** with `/v1` during a deprecation window.

Rationale: path versioning is unambiguous in logs, caches, proxies and SDK base URLs; it is the org convention (HelixVPN `openapi/helix.v1.yaml` is served at `/v1`).

2. What `/v1` covers

`/v1` is the whole authenticated REST surface in `openapi.yaml` **plus** the real-time endpoints in `event-bus-contract.md` (`/v1/events/ws`, `/v1/events/stream`). The unauthenticated ops endpoints (`/healthz`, `/readyz`, `/metrics`) are intentionally **unversioned** — they are operational, not part of the product contract, matching the `helix_proto` decision to keep ops routes out of the versioned slice.

3. Additive vs breaking

Within a major version we make **additive-only** changes. The contract distinguishes:

Non-breaking (allowed within `/v1`, no version bump):

- Add a new endpoint, a new optional request field, or a new response field.
- Add a new enum value **only where the client is documented to tolerate unknowns** (e.g. `Error.code`, `ContentType`, event `type`). SDKs treat unknown enum values as a pass-through/`unknown` variant — see `sdk-strategy.md`.
- Add a new optional query parameter or header.
- Relax a validation constraint; add a new error `code`.

Breaking (requires `/v2`):

- Remove/rename a field, endpoint, or enum value; change a field's type or cardinality.
- Make an optional field required, or tighten validation on existing input.
- Change default behaviour, pagination semantics, or an auth requirement.
- Change the meaning of an existing error `code` or status mapping.

Adding an enum value is treated as **potentially breaking for naive clients**. Because Thready's SDKs are generated with unknown-tolerant enums, it is additive **for conformant SDK users** and called out in the changelog regardless.

4. REST ↔ Protobuf parity

Two schema-first contracts describe one system (see `sdk-strategy.md`):

Plane	Contract	Version token
REST <code>/v1</code>	<code>openapi/thready.v1.yaml</code>	path <code>/v1</code> + <code>info.version</code> semver
Events / DTO wire	<code>proto/helix/thready/v1/</code> <code>*.proto</code>	package <code>helix.thready.v1</code>

The two move together: a change that is breaking in one is breaking in the other, and both bump to `v2` at the same time. Shared DTOs (e.g. the event envelope) are defined once in Protobuf and mirrored in the OpenAPI `components.schemas` so the shapes cannot drift; a contract test asserts the JSON shapes match (see `testing/`).

5. Breaking-change gates (no server CI)

Server-side CI/CD is forbidden `[CONSTITUTION §11.4.156]`, so version discipline is enforced by **local git-hooks + pre-tag full-suite retest** `[CONSTITUTION §11.4.75/40]`, reusing the exact `helix_proto` gates `[VERIFIED: helix_proto Makefile]`:

- **buf breaking** — the proto contract is diffed against the previous tag with `buf breaking --against '.git#tag=<prev>'`; a breaking change inside `v1` **fails the commit** (helix_proto's `buf.yaml` sets `breaking: use: [FILE]`, and its `scripts/buf_breaking_gate.sh` wires it). Renumbering/removing fields inside `v1` is rejected.
- **OpenAPI diff** — `openapi.yaml` is linted (`openapi-lint`) and diffed against the previous tag; removals/renames fail. (`oasdiff`-style check, run locally as part of the Makefile `all` target modelled on helix_proto's `lint → breaking → generate → check-no-handwritten → rust-build → openapi-lint → openapi-generate-ts → roundtrip-test`.)
- **check-no-handwritten** — asserts generated SDK cores were not hand-edited (drift guard), exactly as helix_proto does.

Only after these pass GREEN, a project-prefixed tag `THREADY-<version>` is cut and pushed to all four upstreams `[CONSTITUTION §11.4.151/§2.1]`. The RED-first skeleton that proves the `buf breaking` and OpenAPI-diff gates actually reject a removed field lives in `contract-tests.md` §full-automation.

6. Deprecation & Sunset

When `/v2` ships, `/v1` enters a deprecation window (`[DEFAULT - adjustable] ≥ 6 months`):

- Deprecated operations set `deprecated: true` in OpenAPI and return an RFC 8594 **Sunset: <date>** response header (declared as `components.headers.Sunset`), plus a `Deprecation: true` header.

- The changelog + each SDK release note the deprecation; SDKs emit a one-time warning when a deprecated endpoint is called.
- After the window, `/v1` is removed in a tagged release; the proto `v1` package is likewise retired.

7. SDK & release alignment

- `info.version` in `openapi.yaml` follows **semver**: MINOR for additive, PATCH for fixes, MAJOR only alongside a new URL path segment (`/v2`).
- SDKs are versioned to track the contract's MAJOR.MINOR and published per language registry on each `THREADY-<version>` tag (see `sdk-strategy.md`).
- Tags are mirrored onto every owned submodule and all four upstreams `[CONSTITUTION $11.4.151/$2.1]`; releases carry a changelog + multi-format doc export (Docs Chain).

8. Open items

- `[OPEN: ver-1]` The concrete OpenAPI-diff tool (`oasdiff` vs a custom Go differ) is selected with the tooling pack; the *policy* (additive-only, local gate) is fixed here.
 - `[RESOLVED: ver-2]` The machine-readable `GET /v1/version` endpoint is now specified in `openapi.yaml` (`getVersion` → `VersionInfo`: `api_version`, `contract_hash`, `proto_package`, `build_commit`, `built_at`, and active `deprecations` with `Sunset` dates). It is served under `/v1` for discoverability but is operational in spirit; support quotes `contract_hash` + `build_commit` to pin a client to the exact deployed contract.
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